

1.2 The absolute value

Definition (i) The absolute value of a number x , denoted $|x|$, represents the distance from x to 0 on the real line.

$$(ii) \quad |x| = \begin{cases} x, & \text{if } x \geq 0 \\ -x, & \text{if } x < 0 \end{cases}$$

Example 1 :

$$|3| = \quad ; \quad |-\sqrt{2}| = \quad ; \quad \left|-\frac{5}{4}\right| =$$

Properties : Suppose x and y are any real numbers, and $n \in \mathbb{Z}$.

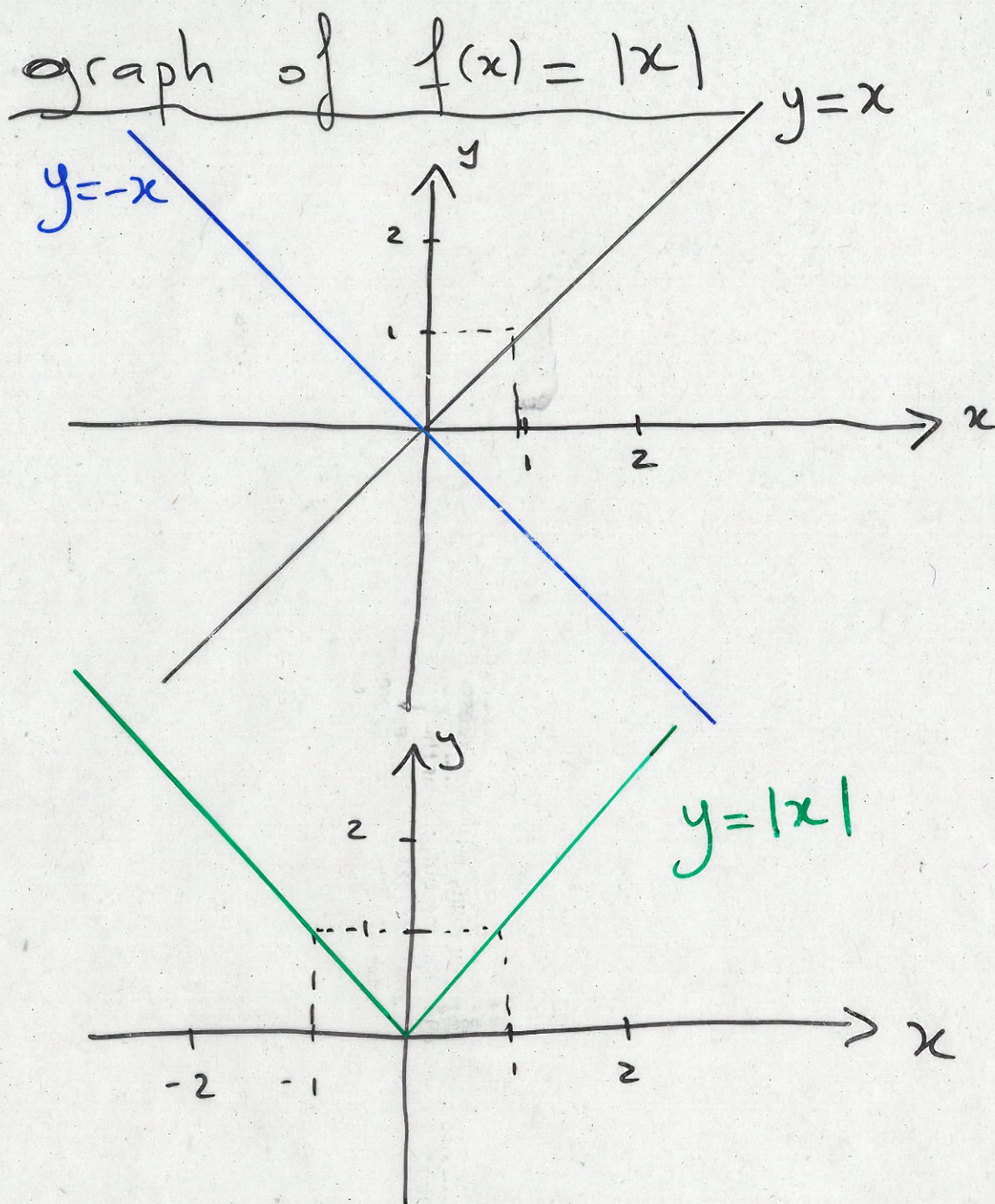
$$(i) \quad |x| \geq 0 \quad \text{for all } x \in \mathbb{R}$$

$$(ii) \quad |x| = 0 \quad \text{if and only if } x = 0$$

$$(iii) \quad |x \cdot y| = |x| \cdot |y|$$

$$(iv) \quad \left|\frac{x}{y}\right| = \frac{|x|}{|y|} \quad (y \neq 0); \quad (v) \quad |x^n| = |x|^n$$

$$(vi) \quad \sqrt{x^2} = |x|$$



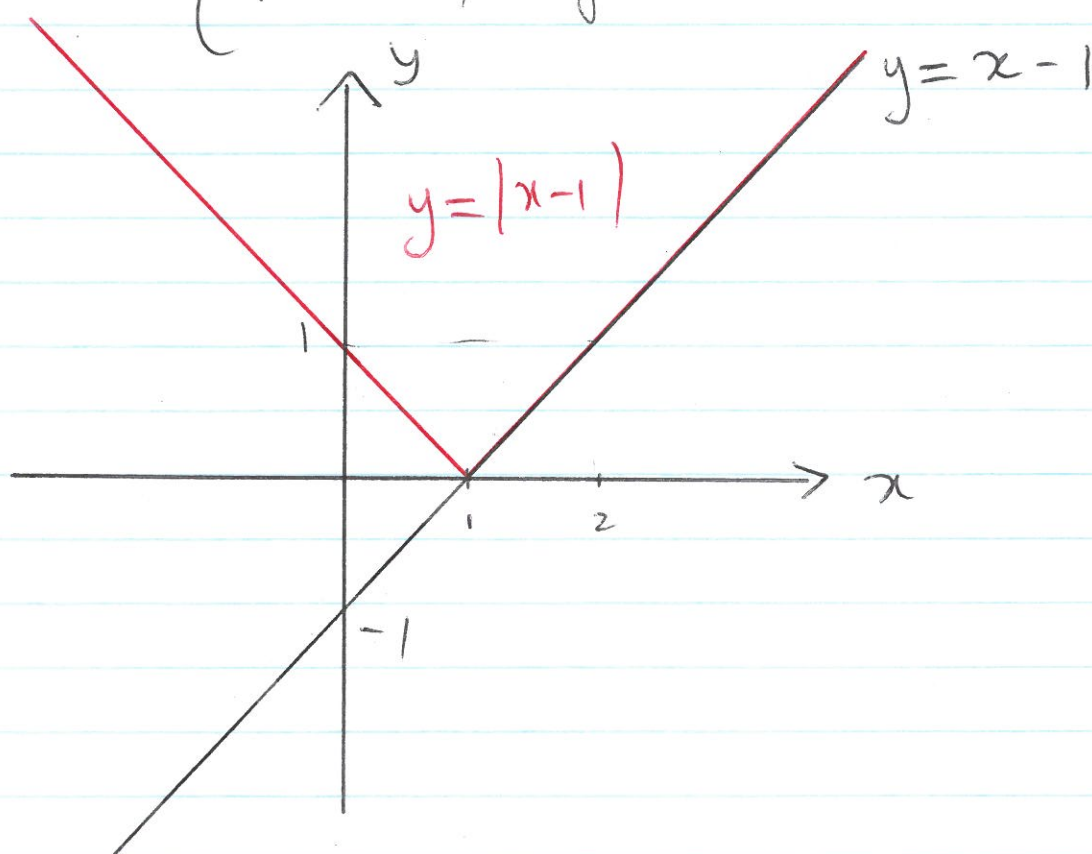
Example 2 : Express $|x-1|$ without using the absolute value symbol and sketch its graph

Example 3 : $g(x) = |x^2 - 1|$

Example 2 :

$$|x-1| = \begin{cases} x-1, & \text{if } x-1 \geq 0 \\ -(x-1), & \text{if } x-1 < 0 \end{cases}$$

$$= \begin{cases} x-1, & \text{if } x \geq 1 \\ 1-x, & \text{if } x < 1 \end{cases}$$



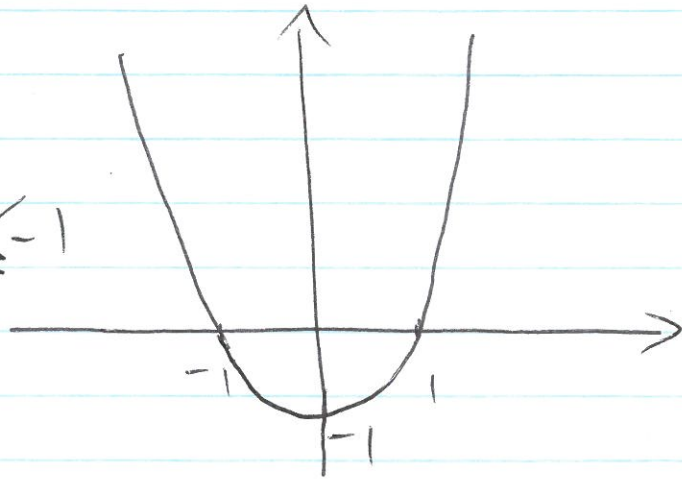
observe that on $(-\infty, 1)$, the symmetry of $y = x-1$ with respect to the x -axis yields $y_2 = 1-x$

Example 3 :

$$|x^2 - 1| = \begin{cases} x^2 - 1, & \text{if } x^2 - 1 \geq 0 \\ -x^2 + 1, & \text{if } x^2 - 1 < 0 \end{cases}$$

$$x^2 - 1 \geq 0$$

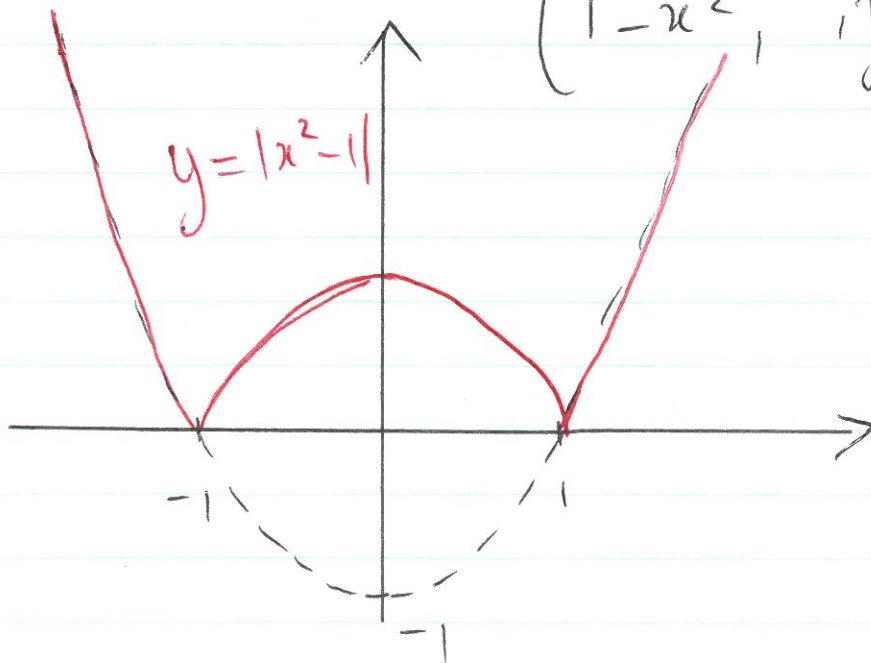
$$\Rightarrow x \geq 1 \text{ or } x \leq -1$$



Hence

$$|x^2 - 1| = \begin{cases} x^2 - 1, & \text{if } x \leq -1 \text{ or } x \geq 1 \\ 1 - x^2, & \text{if } -1 < x < 1 \end{cases}$$

$$= \begin{cases} x^2 - 1, & \text{if } x \in (-\infty, -1] \cup [1, \infty) \\ 1 - x^2, & \text{if } x \in (-1, 1) \end{cases}$$



--- $y = x^2 - 1$
— $y = |x^2 - 1|$